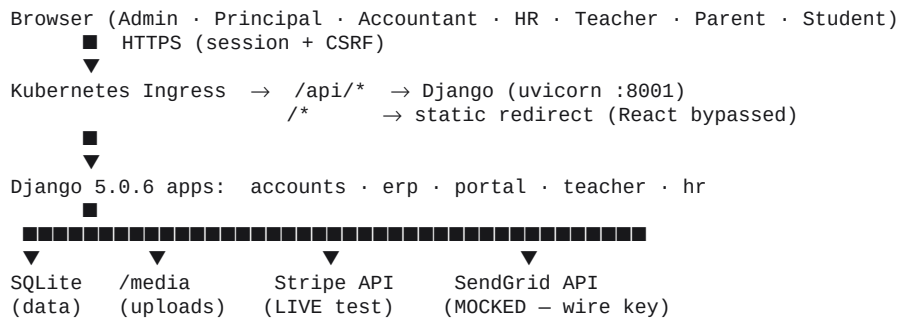




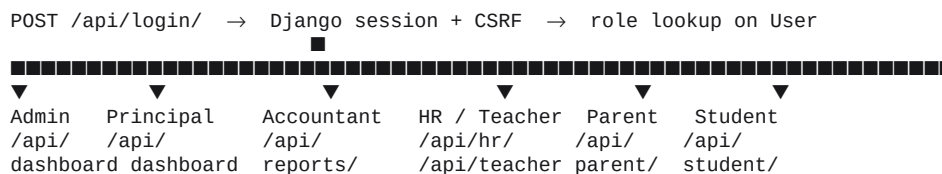
Document purpose: a one-stop reference describing the architecture, user roles, data flows (Input → Process → Output) and operational artefacts of the Instituto São João de Brito Academy & Finance ERP. Suitable for demos, stakeholder review and onboarding new staff or developers.

1 · High-Level Architecture



Stack: Django · Django Templates · Tailwind (locally built) · Chart.js · ReportLab (PDF) · emergentintegrations (Stripe) · SendGrid SDK (gated by env vars).

2 · Authentication & Role Routing

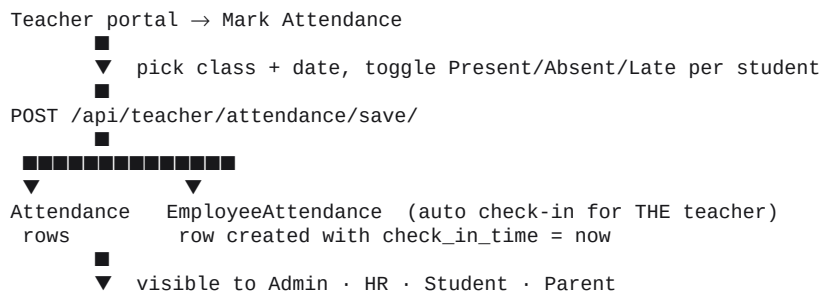


3 · Core Domain — Input · Process · Output

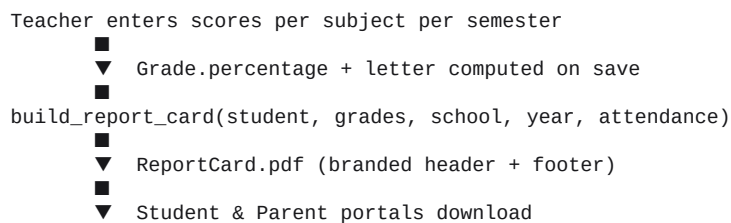
Module	Input	Process	Output
School Profile	Name, motto, address, phone, email, currency, logo (admin form)	School.get_active() updates the singleton row	Branding on every page header + every PDF footer
Students	Admission no., name, DOB, class, parent contacts, ID, photo	Student row created; auto-link to Parent user via email / admission	Detail page, leaderboard, parent & student portal access
Teachers / Employees	Employee no., name, role, subjects, qualifications, salary	Teacher + Employee rows; admin mints teacher login	Teacher list, payroll source, HR headcount KPI
Classes · Subjects · Years	Class name + grade level, subject name + code, year start/end	M2M wiring Teacher↔Subject↔Class↔AcademicYear	Form selects across attendance, grades, report cards

4 · Operational Flows

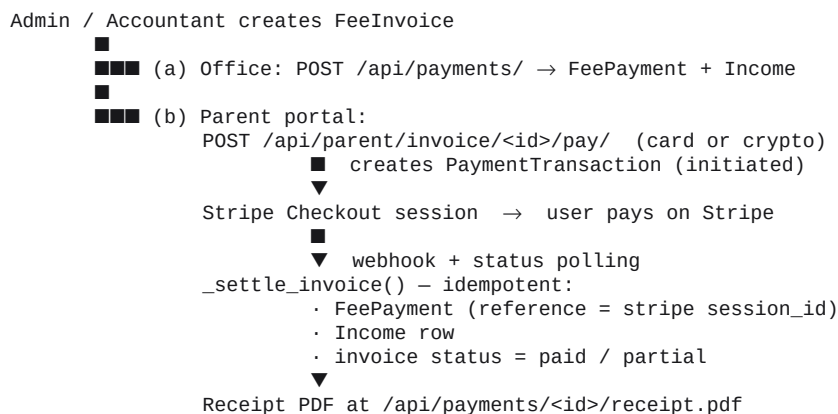
4.1 · Attendance (Teacher action)



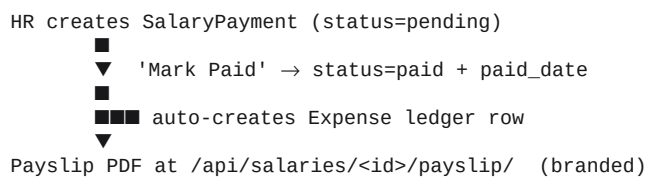
4.2 · Grades & Report Card



4.3 · Fee Invoice → Payment → Income (with online flow)



4.4 · Salary / Payroll



4.5 · Donations · Recruitment · Inventory

Flow	Input	Process	Output
Donations	Donor + amount + date	Auto-mirrors into Income (source=donation)	Donor lifetime total, Finance Report
Recruitment	JobPosting + Candidate stage	Funnel: applied → screened → interview → hired/rejected	HR funnel chart + open-roles KPI
Inventory	Item, qty, unit_cost, min_qty	is_low_stock when $qty \leq min$ · $total_value = qty \times unit_cost$	HR dashboard low-stock list + assignments
Teaching Docs	Admin uploads PDF tagged by Subject (M2M)	Empty M2M = visible to all teachers	Teachers see only docs for their subjects

5 · Reporting Layer

5.1 · Finance Report (/api/reports/finance/)

Helper	Process	Output
_finance_ytd_totals	YTD sums for income, expense, donation, salary, outstanding	5 KPI cards + Net surplus/deficit
_finance_monthly_trend	12-month buckets per stream	Triple-line Chart.js trend
_finance_breakdowns	Group-by category (Expense) and source (Income)	Two doughnut charts

5.2 · HR Dashboard (/api/hr/)

Seven focused helpers — **workforce**, **payroll**, **recruitment**, **training**, **performance**, **today-attendance**, **inventory** — each returns a partial context dict that the view merges. Output: a single dashboard with 18 KPIs + recruitment funnel + low-stock list.

5.3 · Academic Reports

Per-class Report Card ZIP export, plus leaderboards: top student per class (_top_by_class) and per academic year (_top_by_year).

6 · Generated Artefacts

Artefact	Endpoint	Built by	Branded
Report Card	/api/students//report-card/	erp/report_card.py	Yes
Invoice PDF	/api/invoices//pdf/	erp/invoice_pdf.py	Yes
Payment Receipt	/api/payments//receipt.pdf	erp/invoice_pdf.py	Yes
Payslip	/api/salaries//payslip/	erp/payslip_pdf.py	Yes
System Design (this doc)	/api/system-design/pdf/	erp/system_design_pdf.py	Yes

7 · Data Model — Key Relationships

```
User ■■■ role: admin / principal / accountant / hr / teacher / parent / student
■
■■ Teacher (1:1) ■■ Subject (M:N) ■■ Class (M:N) ■■ AcademicYear
■      ■■ SalaryPayment (1:N)
■      ■■ EmployeeAttendance (1:N)
■      ■■ TeachingDocument (1:N)
■
■■ Student (1:1) ■■ Class ████████████████████████████████████████
■      ■■ Attendance (1:N)                                     ■
■      ■■ Grade (1:N) ■■ Subject + AcademicYear ████████
■      ■■ FeeInvoice (1:N) ■■ FeePayment ■■ PaymentTransaction
■      ■■ CreditNote (1:N)
■
■■ Parent (M:N Students)
■
■■ Employee ■■ JobPosting / Candidate / TrainingEnrollment /
           PerformanceReview / LeaveRequest / InventoryAssignment

School (singleton) ■■ currency · logo · address · phone · email · motto
Income / Expense / Donation – ledgers fed by Fee · Salary · Donation flows
```

8 · End-to-End Demo Script

#	Step
1	Login as Admin → Dashboard (10 KPIs, recent activity).
2	Open a Student → profile, grades, invoices, Download Report Card .
3	Finance Report → 12-month trend, doughnut breakdowns, YTD net.
4	HR Dashboard → headcount, payroll, recruitment funnel, low-stock list.
5	Login as Teacher (t-004 / teacher123) → 3 actions only: attendance, grades, evaluations.
6	Mark Attendance for a class → return to Admin → teacher's own HR check-in auto-created.
7	Login as Parent (parent / password123) → see invoices → click Pay (test card 4242 4242 4242 4242).
8	Login as Student (adm-1003 / student123) → read-only grades & invoices.
9	Back to Admin → Salaries → Mark Paid → download branded Payslip PDF.

9 · Live vs MOCKED today

Area	Status	Action to go live
Stripe — Card	LIVE (test keys)	Swap test keys for production Stripe keys
Stripe — Crypto	LIVE	Enable Crypto in Stripe Dashboard
SendGrid email receipts	MOCKED	Set SENDGRID_API_KEY + SENDGRID_FROM_EMAIL in env — no code change
File uploads (/media)	Local disk	Move to S3 / Cloudinary via Django storages
Database	SQLite	Migrate to PostgreSQL for production

10 · Demo Credentials

Role	Username	Password
Admin (superuser)	admin	admin123
Parent (Lucas + Sofía)	parent	password123

Role	Username	Password
Teacher (Carlos Mendes)	t-004	teacher123
Student (Miguel Costa)	adm-1003	student123
Accountant	accountant	password123
HR Manager	hr	password123
Principal	principal	password123

Stripe test card: **4242 4242 4242 4242** · any future expiry · any CVC.